Qi Xu

PhD Student in Artificial Intelligence @ Westlake University

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EDUCATION

Westlake University — 2026 – Present
PhD Student in Artificial Intelligence
Supervised by Peidong Liu; Research: Embodied AI, Multimodal Large Language Models

Wuhan University — 2023 – 2026
Master in Pattern Recognition and Intelligent Systems
Research: Computer Vision, 3D Reconstruction, Multimodal Learning

Wuhan University — 2019 – 2023
B.S. in Spatial Information and Digital Technology

EXPERIENCE

ByteDance, TikTok Live – LLM Applications Team — 2025 – 2026
Research Intern

- Developed and optimized **Multi-Modal Large Language Models (MLLMs)** for temporal grounding tasks using SFT and Reinforcement Learning (RL) pipelines.
- Built scalable data pipelines for massive MLLM training, enabling efficient collection and annotation of high-quality temporal grounding datasets.
- Conducted **large-scale distributed training on hundreds of GPUs**, optimizing training efficiency and model convergence for production-level deployment.
- Designed agentic workflows leveraging LLMs to automate real-time interaction analysis and optimize content distribution within the TikTok Live ecosystem.

SELECTED PUBLICATIONS

SIU3R: Simultaneous Scene Understanding and 3D Reconstruction Beyond Feature Alignment — 2025
NeurIPS 25 Spotlight Project Page ArXiv Code
Qi Xu*, Dongxu Wei*, Lingzhe Zhao, Wenpu Li, Zhangchi Huang, Shunping Ji, Peidong Liu
Proposed an alignment-free framework achieving SOTA 3D reconstruction and scene understanding from unposed images via shared pixel-aligned 3D representation.

Towards One-to-Many Temporal Grounding — 2026
ICML 26 Project Page ArXiv
Qi Xu*, Yue Tan*, Shihao Chen, Jiahao Meng, Anran Wang, Shunping Ji, Hao Fei, Xiangtai Li
Introduced the first systematic solution for OMTG, featuring a 56k-sample dataset and RL training with CoT-based rewards; outperformed Gemini 2.5 Pro by 15.85%.

Any 3D Scene is Worth 1K Tokens: 3D-Grounded Representation for Scene Generation at Scale — 2026
NeurIPS Submission Project Page ArXiv Code
Dongxu Wei*, **Qi Xu***, Zhiqi Li, Hangning Zhou, Cong Qiu, Hailong Qin, Mu Yang, Zhaopeng Cui, Peidong Liu
First approach to perform 3D scene generation directly within an implicit 3D latent space using 3D Diffusion Transformers.

HiCI: Hierarchical Construction–Integration for Long-Context Attention — 2026
ICML 26 ArXiv Code
Xiangyu Zeng, **Qi Xu**, Yunke Wang, Chang Xu
Efficient long-context modeling using hierarchical attention, extending LLaMA-2 to 100K tokens with only 5.5% additional parameters.

Watch, Remember, Reason: Human-View Video Understanding with MLLMs — 2026
arXiv 26 ArXiv GitHub
Jiahao Meng, Yue Tan, **Qi Xu**, Kuan Gao, Weisong Liu, Yanwei Li, Jason Li, Lingdong Kong, Haochen Wang, Qianyu Zhou, Jiangning Zhang, Guangliang Cheng, Yunhai Tong, Lu Qi, Minghsuan Yang
Surveyed human-view video understanding with MLLMs through watching, remembering, and reasoning abilities for long, multimodal, and knowledge-intensive video scenarios.

VideoZeroBench: Probing the Limits of Video MLLMs with Spatio-Temporal Evidence Verification — 2026
NeurIPS Submission Project Page ArXiv Code
Jiahao Meng, Tan Yue, **Qi Xu**, Haochen Wang, Zhongwei Ren, Weisong Liu, Yuhao Wang, Renrui Zhang, Yunhai Tong, Haodong Duan
A hierarchical benchmark exposing the gap between surface-level answer correctness and genuine evidence-based reasoning in long-video understanding.

E-MoFlow: Learning Egomotion and Optical Flow from Event Data via Implicit Regularization
NeurIPS 25 Project Page ArXiv Code
Wenpu Li, Bangyan Liao, Yi Zhou, **Qi Xu**, Pian Wan, Peidong Liu — 2025
An unsupervised framework for joint ego-motion and optical flow estimation using implicit neural representations and differential geometric constraints.

PUBLICATIONS WITH ADVISED STUDENTS

SupIR-GS: Thermal Infrared Super-Resolution Novel View Synthesis with Imaging-Calibrated 3DGS
ECCV 26 Corresponding Author
Jin Liu, Haodong Li, Jiagang Chen, Dabin leng, Jiguang Li, Zhao Huang, Xiaoshuai Zhang, **Qi Xu**†, Zhiwen Zheng, Xingru Huang† — 2026
Reconstructs high-resolution 3D thermal scenes from low-resolution inputs via physics-informed degradation modeling.

Physically Grounded Dual-Opacity Gaussian Splatting for Joint RGB-TIR Reconstruction
ECCV 26 Corresponding Author
Jin Liu, Dabin leng, Jiagang Chen, Haodong Li, Jiguang Li, Zhao Huang, Xiaoshuai Zhang, Zhiwen Zheng, Xingru Huang†, **Qi Xu**† — 2026
A unified RGB-TIR reconstruction framework resolving cross-spectral visibility conflicts through thermal field modeling.

TECHNICAL SKILLS

Deep Learning: PyTorch, Transformers, MLLMs, Diffusion Models, Reinforcement Learning
Engineering: Ray, Distributed Training (Multi-GPU/Cluster), Linux, Data Pipelines
Research Interests: Embodied AI, Multimodal Large Language Models, 3D Foundation Models
Languages: English (CET6), Mandarin (Native)